

James Hill

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Professional Summary

Software engineer with experience building and supporting production enterprise systems using Java, C#, SQL, integration middleware, and distributed messaging technologies. Experienced in backend development, workflow/state-machine design, database programming, automated testing, debugging, and delivery in large Agile environments. Strong foundation in object-oriented design, concurrency, REST APIs, containers, and cloud-native development.

Technical Skills

Languages: Java, C#, C/C++, SQL, JavaScript/TypeScript, Python, Ruby, Bash

Backend & Data: REST APIs, DB2, IBM ACE/IIB, IBM MQ, IBM RSA, WTX

Testing: JUnit, Mockito, Cucumber, Citrus

Infrastructure & Tools: Docker, Kubernetes, AWS, Git, Linux

Web: React, Next.js, Angular

Relevant Experience

Scotiabank

Software Engineer II

March 2026 – Present

Toronto, Ontario

- Develop and support high-volume enterprise transaction-processing systems for domestic and international services, with emphasis on reliability, data integrity, and maintainability.
- Build backend integration logic and message-processing workflows using IBM ACE/IIB, IBM FTM, IBM RSA, IBM MQ, DB2.
- Design and implement finite-state machines (FSMs) that coordinate complex asynchronous workflows, event handling, validation, routing, and system state transitions.
- Write and optimize SQL/DB2 operations for transactional workloads, including queries, updates, multi-row processing, transaction boundaries, locking, and concurrency-sensitive logic.
- Develop integration components for message transformation, validation, database access, routing, and communication between distributed enterprise services.
- Diagnose production and integration issues across application, messaging, workflow, and database layers; analyze race conditions, transaction behavior, event sequencing, and system logs.
- Contribute to automated integration and behavior-driven testing using Cucumber and Citrus, alongside developer testing and regression validation.
- Collaborate with architects, developers, QA, and business stakeholders in an Agile environment to deliver changes to business-critical systems.

FDM Group

Software Engineering Consultant

September 2025 – Present

Toronto, Ontario

- Developed backend and web applications using Java, C#, SQL, REST APIs, and modern object-oriented development practices.
- Applied SOLID principles, common design patterns, multithreading, lambdas/streams, structured logging, and static-analysis tooling to build maintainable software.
- Built and tested application components using JUnit and Mockito while working in Agile/Scrum delivery environments.
- Worked with relational databases, containerized development environments, version control, and CI-oriented engineering workflows.

DataAnnotation Tech

Software Developer – AI Trainer (Contract)

January 2024 – Present

Remote

- Evaluate, debug, and improve AI-generated software solutions across Java, C#, C/C++, Python, JavaScript, Rust, Kotlin, and Bash.
- Train and evaluate AI coding agents by reviewing, correcting, and improving step-by-step problem-solving workflows, tool usage, code changes, and implementation decisions.
- Contributed to training and evaluation work involving Claude, Grok, and Gemini coding models.
- Review code for correctness, instruction adherence, API usage, edge cases, and software-engineering quality; perform QA on other technical analysts' work.
- Assess solutions involving cloud and web technologies including AWS, Azure, Next.js, pandas, and third-party APIs.

- Developed and maintained application features using JavaScript, C#, Java, Angular, Ruby, SQLite, Docker, Kubernetes.
- Collaborated across development teams to diagnose defects, implement fixes, and deliver new builds in an Agile environment.
- Created Grunt automation tasks for cache-control workflows and used Git/TFS for collaborative version control.

Projects

- Pluribus-Inspired Poker AI** *C++ / C*
- Built a poker engine using object-oriented design and implemented a Counterfactual Regret Minimization based AI.
 - Designed a thread-safe Monte Carlo CFR training model for generating a blueprint strategy, applying concurrency and algorithmic optimization techniques.

- Banking REST API** *C# / .NET / SQL*
- Built a .NET backend with JWT authentication, role-based authorization, CQRS-style handlers, repositories, and stored-procedure-based persistence.
 - Implemented customer, account, transaction, user, and refresh-token workflows with automated tests using xUnit, Moq.

- Ocean Change App** *Flutter / Dart / Firebase*
- Developed a mobile application for reporting environmental observations using Flutter, Firebase/Firestore, Hive, Xcode, and TestFlight.

- Portfolio Website** *Next.js / React / Tailwind CSS*
- Built and deployed a responsive developer portfolio with Next.js and React on Vercel.

Education

- Oregon State University** **August 2024**
Bachelor of Science in Computer Science (Postbaccalaureate), GPA: 4.0
- University of Ottawa** **April 2020**
B.Sc. Honours Biochemistry, Co-op – Magna Cum Laude *Ottawa, Ontario*

Languages

English (Fluent) | French (Proficient)